

HW2: Transformer Language Modeling

Nicholas McCracken
mccracken.161@osu.edu

1 Part 1: Transformer Encoder for Letter Counting

A simplified Transformer encoder was implemented from scratch using character embeddings and positional encoding to classify each position in the string based on how many times the character at that position occurred before. A Transformer Layer was built with a self-attention layer followed by a feedforward network.

Table 1 summarizes the convergence behavior between variants of this Transformer encoder model. A single-layer Transformer reaches the 0.99 dev accuracy threshold by epoch 9. By increasing the depth to three layers, the convergence is accelerated, reaching the same threshold by epoch 5, although the final performance of both models remains nearly identical after convergence. This implies that additional transformer layers primarily improve the optimization rather than the capacity of the model for this letter counting task.

Overall, the task is simple enough so that this shallow network is adequate for our purposes.

2 Exploration 1: Attention Visualization

Figures 1 and 2 show attention maps from a single-layer and three-layer Transformer using base attention for a single sentence. In Figure 1, the single-layer model shows strong attention activations at previous occurrences of the same character. Repeated characters such as "i" and "n" receive very high attention at later positions.

Figure 2 shows attention from the deeper three-layer model. While the model still captures these repeated characters, the attention pattern diffuses more in the lower triangle of the map. This implies that the deeper transformer struggles to meaningfully improve since this task is on such a short sentence of characters. On a larger sentence, the attention map might be more focused.

3 Exploration 2: Relative Position Attention

Relative position attention (Shaw et al., 2018) was implemented by incorporating adding scores based on relative offsets between character positions to the Query * Key similarity score. This extra term contributes additional position-dependent data to the attention scores.

As shown in Table 1, the relative position model converges at a similar rate to the single-layer base attention model. Figure 3 visualizes its attention map. Compared to Figure 1, the relative position model puts stronger emphasis on nearby tokens rather than reoccurring tokens. This reflects the explicitness of the offset values, reflecting its explicit modeling of positional offsets. In spite of this change, the convergence speed and final accuracy remain almost unchanged.

Because the maximum sequence length is only 20 characters, the positional encodings already provide more than sufficient positional information. On larger sentences, the relative position model would have more noticeable differences to the base attention model.

4 Part 2: Transformer Language Model

A Transformer language model was implemented using nn.TransformerEncoder with masking to prevent attention to future tokens. The model processes fixed length chunks of 20 characters and predicted the next character at every position simultaneously based on previous characters.

The final model achieved a perplexity of 5.7 on the development set after 15 epochs. Even after 15 epochs, the model was continuing to improve and had not yet converged.

This shows that even a shallow Transformer network can be effective when modeling character dependencies. This type of architecture could be used in a LLM to generate strings as well.

Model	Converged Epoch
1-Layer Transformer (Base Attention)	9
3-Layer Transformer (Base Attention)	5
1-Layer Transformer (Relative Position)	9

Table 1: Convergence comparison for the letter counting task. Converged epoch is defined the first epoch where dev accuracy ≥ 0.99 .

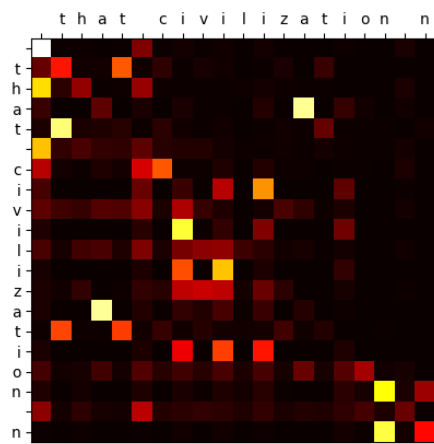


Figure 1: 1-layer Transformer with base attention.

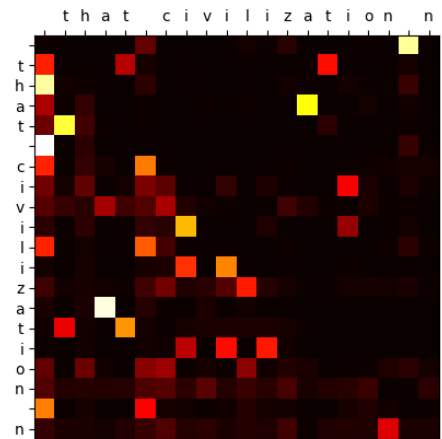


Figure 3: 1-layer Transformer with relative position attention.

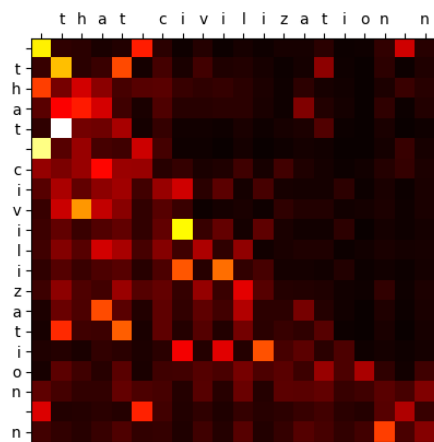


Figure 2: 3-layer Transformer with base attention.